

Windows CMD Essentials: Lost & Found — Instructor Guide

Every challenge, answer, and accepted variant · generated 2026-08-25 from the live pack data

Quick reference

Set your own site URL and class password before handing this out. The admin handle is whatever you put in `ADMIN_HANDLES` — `instructor`, say; there is no default, and the setting ships empty.

Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**.

30 challenges · 655 points total · finds look like `FIND{ABCD2345EFGH}` and are **different for every student**.

How the difficulty fades — Act I briefs show the whole command, and the free hint repeats it. From Act II the brief names the tool but not its flags or arguments, and the exact line moves into a hint that costs XP. By the last act the brief states only the objective. A student who never buys a hint has genuinely recalled every command. Hint spend is visible in the instructor dashboard, so it reads as a difficulty signal rather than as cheating.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to click the green flag in the terminal to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. <code>cd ~</code> returns home.
"It says command not found"	If it's a real command we don't simulate (<code>top</code> , <code>nano</code>), the game says so explicitly. Genuine typos say "command not found".
"I don't know which command to use"	Intended from Act II on. Point them at the free hint first, then the Reference tab. The costed hint is the safety net, not a failure.
"The act is locked"	Each act opens after solving all but one of the previous act's challenges.
"Nothing printed"	Correct for <code>cd</code> and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: Whose Machine Is This?	Walk the disk with <code>CD</code> , <code>DIR</code> switches, <code>TYPE</code> , <code>TREE</code> and <code>WHERE</code>	11	180	Always open
Act II: Tidy It Up	Copy, move, rename, delete, and read the machine's own environment	11	240	Solve 9 of 11 in the prior act

Act III: Fill In the Form	FINDSTR, pipes, diagnostics, and a hash for the property record	8	235	Solve 9 of 11 in the prior act
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Students may skip **one** challenge per act and still advance.

Act I: Whose Machine Is This?

Walk the disk with CD, DIR switches, TYPE, TREE and WHERE

Where Am I? (CD)

10 pts · w1-cd-pwd · OK

Task: The laptop is already open at a Command Prompt. In Windows CMD, running `cd` with no arguments prints the directory you are in — run it and see whose profile you have landed in.

Answer: `cd`

Completion: Auto-completes on: allOf

Hint 1 (free): Type `cd` and press Enter.

Also accepted: `chdir`

Directory Listing (DIR)

10 pts · w1-dir · OK

Task: Run `dir` to list the files and folders of the profile you have landed in.

Answer: `dir`

Completion: Auto-completes on: allOf

Hint 1 (free): Type `dir`.

Also accepted: `dir .`

Show All Files (DIR /A)

15 pts · w1-dir-a · OK

Task: One file in this folder never turns up in a plain `dir`: Windows hides a file with an attribute letter (H for Hidden) rather than with a leading dot. Make `dir` show everything, hidden files included.

Answer: `dir /a`

Completion: Auto-completes on: allOf

Hint 1 (free): `dir` switches start with a forward slash. The one for attributes is `/a`.

Hint 2 (-5 XP): Run `dir /a`.

Also accepted: `dir /a:h`, `dir /a /b`

Bare Listing (DIR /B)

15 pts · w1-dir-b · OK

Task: Produce a clean listing of this folder: names only, with no header, no sizes and no summary line. Use `dir`.

Answer: `dir /b`

Completion: Auto-completes on: allOf

Hint 1 (free): The `dir` switch for a bare listing is `/b`.

Hint 2 (-5 XP): Run `dir /b`.

Also accepted: `dir /b .`

Recursive Search (DIR /S)

20 pts · w1-dir-s · OK

Task: List Documents AND every folder inside it in one go. Use `dir` .

Answer: `dir /s Documents`

Completion: Auto-completes on: allOf

Hint 1 (free): The `dir` switch that walks down into subfolders is `/s` .

Hint 2 (-5 XP): Run `dir /s Documents` .

Also accepted: `dir /s` , `dir /s .`

Enter Documents (CD)

15 pts · w1-cd-nav · OK

Task: The owner's paperwork is in Documents . Run `cd Documents` to move into it.

Answer: `cd Documents`

Completion: Auto-completes on: cwdIs

Hint 1 (free): Type `cd Documents` .

Also accepted: `cd .\Documents` , `cd C:\Users\Student\Documents`

Climb to Parent (CD ..)

15 pts · w1-cd-parent · OK

Task: Run `cd ..` from Documents to climb back up to the profile folder.

Answer: `cd ..`

Completion: Auto-completes on: cwdIs

Hint 1 (free): Type `cd ..` .

Also accepted: `cd C:\Users\Student`

Display File Content (TYPE)

15 pts · w1-type · OK

Task: Print Documents\notes.txt to the prompt and find out what domain this laptop belongs to. Windows has no `cat` .

Answer: `type Documents\notes.txt`

Completion: Auto-completes on: outputEquals

Hint 1 (free): The cmd.exe command that prints a file is `type` .

Hint 2 (-5 XP): Run `type Documents\notes.txt` .

Also accepted: `type Documents/notes.txt`

Directory Tree (TREE)

15 pts · w1-tree · OK

Task: Show this folder and everything under it as a picture of the hierarchy rather than a flat list.

Answer: `tree`

Completion: Auto-completes on: allOf

Hint 1 (free): The command is `tree` . Adding `/f` lists the files as well as the folders.

Hint 2 (-5 XP): Run `tree` .

Also accepted: `tree /f`

Locate Binaries (WHERE)

15 pts · w1-where · OK

Task: Find out where on this disk the command interpreter itself lives — the program `cmd` .

Answer: `where cmd`

Completion: Auto-completes on: `allOf`

Hint 1 (free): `where` searches the folders on PATH for a named program, the way `which` does on Linux.

Hint 2 (-5 XP): Run `where cmd` .

Also accepted: `where cmd.exe`

BOSS: The Four Servers

35 pts · w1-boss · OK

Task: Somewhere under this profile is a file listing four servers and one recovery code. Nobody will tell you where it is: walk the disk until you find it, print it, and submit the recovery code — it was generated for you alone.

Answer: `type Documents\servers.txt`

Completion: Submits a flag · flag is inside `C:\Users\Student\Documents\servers.txt`

Hint 1 (free): A recursive listing shows every file on the machine at once. Both `dir` and `tree` can do that with the right switch.

Hint 2 (-10 XP): The file is `Documents\servers.txt` . Print it with `type` , then paste the `FIND{...}` value into the box.

Also accepted: `type Documents/servers.txt` , `type C:\Users\Student\Documents\servers.txt`

Act II: Tidy It Up

Copy, move, rename, delete, and read the machine's own environment

Copy Files (COPY)

20 pts · w2-copy · OK

Task: Never edit the only copy of anything. Make a duplicate of `Documents\notes.txt` called `Documents\notes_backup.txt`.

Answer: `copy Documents\notes.txt Documents\notes_backup.txt`

Completion: Auto-completes on: `allOf`

Hint 1 (free): One `cmd.exe` command duplicates a file without touching the original. It takes the existing file first, then the path for the copy.

Hint 2 (-10 XP): Run `copy Documents\notes.txt Documents\notes_backup.txt`.

Also accepted: `copy Documents/notes.txt Documents/notes_backup.txt`

Move Files (MOVE)

20 pts · w2-move · OK

Task: The owner left a to-do list loose on the Desktop. Move `Desktop\todo.txt` into the `Documents` folder, keeping the same filename.

Answer: `move Desktop\todo.txt Documents\todo.txt`

Completion: Auto-completes on: `allOf`

Hint 1 (free): A different command relocates a file instead of duplicating it — same two-argument order as before, but nothing survives at the old path afterward.

Hint 2 (-10 XP): Run `move Desktop\todo.txt Documents\todo.txt`.

Also accepted: `move Desktop\todo.txt Documents`

Rename Files (REN)

20 pts · w2-ren · OK

Task: Rename `Desktop\Shortcut.lnk` to `AppShortcut.lnk`. It must stay on the Desktop.

Answer: `ren Desktop\Shortcut.lnk AppShortcut.lnk`

Completion: Auto-completes on: `fileExists`

Hint 1 (free): Renaming is a command of its own, separate from moving a file. Give it the existing path, then only the new name — a bare name, not a second full path, since the file is not going anywhere.

Hint 2 (-10 XP): Run `ren Desktop\Shortcut.lnk AppShortcut.lnk`.

Also accepted: `rename Desktop\Shortcut.lnk AppShortcut.lnk`

Delete Files (DEL)

20 pts · w2-del · OK

Task: Downloads\archive.zip is junk the owner will not want back. Delete it — and on the same line list Downloads again so you can see it has gone. In cmd.exe, & runs one command after another.

Answer: del Downloads\archive.zip & dir /b Downloads

Completion: Auto-completes on: allOf

Hint 1 (free): Deleting a file at the cmd.exe prompt takes one command and a path — no confirmation dialog, no Recycle Bin. Chain a listing after it with & so you watch the file actually disappear.

Hint 2 (–10 XP): Run del Downloads\archive.zip & dir /b Downloads.

Also accepted: erase Downloads\archive.zip & dir /b Downloads

Create Directories (MD / MKDIR)

20 pts · w2-md · OK

Task: Create a folder called Backup in the profile folder, to hold what you are about to put aside.

Answer: md Backup

Completion: Auto-completes on: dirExists

Hint 1 (free): cmd.exe has a short spelling and a long spelling for the same command that creates a folder. Give it the name you want.

Hint 2 (–10 XP): Run md Backup.

Also accepted: mkdir Backup

Remove Directories (RD / RMDIR)

20 pts · w2-rd · OK

Task: The owner left an empty Projects folder behind and it is not going back with the laptop. Remove it, and on the same line list the profile folder again so you can see it has gone.

Answer: rd Projects & dir /b

Completion: Auto-completes on: allOf

Hint 1 (free): The command that removes a folder also comes in a short and a long spelling, and refuses to run if anything is still inside. Chain a listing after it with & to see the folder gone.

Hint 2 (–10 XP): Run rd Projects & dir /b.

Also accepted: rmdir Projects & dir /b

Inspect File Attributes (ATTRIB)

25 pts · w2-attrib · OK

Task: Documents\hidden_config.ini does not appear in a plain dir either. Show which attribute letters are set on it.

Answer: attrib Documents\hidden_config.ini

Completion: Auto-completes on: allOf

Hint 1 (free): One command reports a file's attribute letters without changing anything — give it the path and read what comes back: H is Hidden, R is Read-only, S is System, A is Archive.

Hint 2 (–10 XP): Run attrib Documents\hidden_config.ini.

Also accepted: attrib Documents/hidden_config.ini

Environment Variables (%VAR%)

20 pts · w2-env-var · OK

Task: cmd.exe wraps an environment variable in percent signs, like %NAME%. Print the value that USERPROFILE holds on this machine.

Answer: `echo %USERPROFILE%`

Completion: Auto-completes on: allOf

Hint 1 (free): One command just prints whatever text it is given. cmd.exe swaps %VAR% for its value before that command ever runs, so it never even sees a variable name.

Hint 2 (-10 XP): Run `echo %USERPROFILE%`.

Also accepted: `echo %userprofile%`

Setting Variables (SET)

25 pts · w2-set · OK

Task: You just read %USERPROFILE% back from the environment — now put something of your own into it. Every later write into Documents should trace back to this laptop's case, so set a marker before you copy anything else: MY_VAR holding 123, printed back on the same line to prove it took. set prints nothing when it works, and & joins two commands.

Answer: `set MY_VAR=123 & echo %MY_VAR%`

Completion: Auto-completes on: allOf

Hint 1 (free): One new command creates the variable with no spaces around the equals sign. echo prints it back the same way you just used, inside percent signs, joined on the same line with &.

Hint 2 (-10 XP): Run `set MY_VAR=123 & echo %MY_VAR%`.

Also accepted: `set MY_VAR=123 && echo %MY_VAR%`

Clear the Console (CLS)

10 pts · w2-cls · OK

Task: The screen is cluttered with everything you have run so far. Clear it.

Answer: `cls`

Completion: Auto-completes on: allOf

Hint 1 (free): cmd.exe has one command for wiping the screen, and it takes nothing after it — no path, no switch, just the word on its own.

Hint 2 (-10 XP): Run `cls`.

BOSS: Ready for Collection

40 pts · w2-boss · OK

Task: Put the paperwork together for the front desk. Make a folder called Return, copy Documents\notes.txt into it, and list that folder to prove the copy landed — all on one line, joined with &.

Answer: `md Return & copy Documents\notes.txt Return\notes.txt & dir /b Return`

Completion: Auto-completes on: allOf

Hint 1 (free): Three commands in a row on one line: make the directory, copy the file into it, then list what is inside it to prove it landed.

Hint 2 (-15 XP): Run `md Return & copy Documents\notes.txt Return\notes.txt & dir /b Return`.

Also accepted: `mkdir Return & copy Documents\notes.txt Return\notes.txt & dir /b Return`

Act III: Fill In the Form

FINDSTR, pipes, diagnostics, and a hash for the property record

Pattern Search (FINDSTR)

25 pts · w3-findstr · OK

Task: The property form asks what this machine was doing. Show every line in `logs\eventlog.txt` about a new process being created. Windows has no `grep`.

Answer: `findstr "A new process" logs\eventlog.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): `cmd.exe`'s tool for searching inside a file takes the phrase to look for first, then the file to search — quote the phrase, since this one has spaces in it.

Hint 2 (–10 XP): Run `findstr "A new process" logs\eventlog.txt`.

Also accepted: `findstr /c:"A new process" logs\eventlog.txt`, `type logs\eventlog.txt | findstr "A new process"`

Case-Insensitive FINDSTR (/I)

25 pts · w3-findstr-i · OK

Task: The form asks whether the machine's protection was switched on. The event log names the product inconsistently — sometimes capitalised, sometimes not — so a search that matches one spelling misses the rest.

Answer: `findstr /i defender logs\eventlog.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): `findstr` switches start with a forward slash. The one that ignores case is `/i`.

Hint 2 (–10 XP): Run `findstr /i defender logs\eventlog.txt`.

Also accepted: `findstr /i "defender" logs\eventlog.txt`, `findstr /I defender logs\eventlog.txt`

Count Lines (FIND /C /V "")

30 pts · w3-find-count · OK

Task: Count how many lines `logs\firewall.log` contains. `cmd.exe` has no `wc`, so the standard trick is to print the file and pipe it into a counter that counts every line NOT matching an empty pattern.

Answer: `type logs\firewall.log | find /c /v ""`

Completion: Auto-completes on: allOf

Hint 1 (free): `cmd.exe`'s line-counting trick reuses a search tool: point it at a pattern so broad that every single line matches, and ask it to report a count of matches instead of the matching text.

Hint 2 (–10 XP): Run `type logs\firewall.log | find /c /v ""`.

Instructor note: The `find /c /v ""` idiom is genuinely unguessable — expect most of the class to buy the hint.

Process List (TASKLIST)

25 pts · w3-tasklist · OK

Task: Record what is running on the machine before you shut it down. Show every process, with its memory use.

Answer: `tasklist`

Completion: Auto-completes on: allOf

Hint 1 (free): cmd.exe's equivalent of Task Manager takes nothing after it — no path, no switch, just the command on its own — and lists every running process with its memory use.

Hint 2 (–10 XP): Run `tasklist`.

Network Configuration (IPCONFIG)

25 pts · w3-ipconfig · OK

Task: The form has a line for the network the laptop was last on. Show this machine's IPv4 address and adapter configuration.

Answer: `ipconfig`

Completion: Auto-completes on: allOf

Hint 1 (free): One cmd.exe command, run with nothing after it, reports every network adapter on this machine and the address each one holds.

Hint 2 (–10 XP): Run `ipconfig`.

Also accepted: `ipconfig /all`

System Information (SYSTEMINFO)

25 pts · w3-systeminfo · OK

Task: The form wants the operating system build and the installed memory. Display the machine's system summary.

Answer: `systeminfo`

Completion: Auto-completes on: allOf

Hint 1 (free): One cmd.exe command, run with nothing after it, dumps the machine's whole system summary — OS build, installed memory, all of it.

Hint 2 (–10 XP): Run `systeminfo`.

Cryptographic Hash (CERTUTIL)

35 pts · w3-certutil · OK

Task: The form wants a checksum of `Documents\data.csv`, so that nobody can alter it between now and collection without it showing. Compute its MD5. Windows has no `md5sum`; the built-in tool is `certutil`.

Answer: `certutil -hashfile Documents\data.csv MD5`

Completion: Auto-completes on: allOf

Hint 1 (free): One cmd.exe tool computes a file's hash. It needs a switch that names the file to hash, then the path, then the algorithm name in capitals.

Hint 2 (–10 XP): Run `certutil -hashfile Documents\data.csv MD5`.

Also accepted: `certutil -hashfile Documents/data.csv MD5`

BOSS: Name on the Ticket

45 pts · w3-boss · OK

Task: The last line of the form is the one that matters: which account logged on to this machine successfully. Windows records that as event 4624 in `logs\eventlog.txt`. Print exactly that one line, and nothing else.

Answer: `findstr 4624 logs\eventlog.txt`

Completion: Auto-completes on: `outputEquals`

Hint 1 (free): Two ways in, and both work: search the log for the event number, or search it for the words that describe a successful logon.

Hint 2 (-15 XP): Run `findstr 4624 logs\eventlog.txt`.

Also accepted: `findstr /c:"EVENT ID 4624" logs\eventlog.txt`, `type logs\eventlog.txt | findstr 4624`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes around a path
- Path prefixes: `./Documents`, `~/Documents`, or the full absolute path
- Short paths when already inside the folder
- Flag spacing: `head -n 5`, `head -n5`, `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent. Point stuck students there before they buy a hint.
- **Unsimulated tools** reply with what they are, rather than a bare "command not found".
- **Hints:** the first is free and conceptual; the one that gives the exact command costs XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.
- **Watch the stuck points.** The dashboard flags any challenge with a solve rate under 35%. Now that briefs no longer contain the answer, that number means something: it marks a concept the class did not get, not a typing error.